COM5231 · Introduction to Wireless and Ad Hoc Networking

# Question-Based Summary

Every past-paper question with an exam-ready answer, plus likely questions, formulas and common mistakes

SUPPLEMENTARY content not found in the uploaded lecture notes

NOTES ERROR a mistake in the lecture notes, with the verified correction

EXAM how the topic is or could be examined

★ high-priority, frequently examined content

Exam preparation pack, built from the uploaded lecture materials and past papers

# Contents

How to Use This Summary

1. Wireless Channel and Propagation

2. Channel Correction, Capacity and MIMO

3. Multiple Access and CDMA

4. Spread Spectrum

5. OFDM and OFDMA

6. Error Detection and Correction

7. Wireless LANs (IEEE 802.11)

8. Bluetooth and IEEE 802.15

9. Routing and Ad Hoc Networks (SUPPLEMENTARY: no notes provided)

10. One-Page Formula and Method Sheet

# How to Use This Summary

This is the **whole module condensed into questions and answers**. Every past-paper question from all five papers (Exam, Test 1A, Test 1B, Test 2 2025, Test 2 2026) is here, plus the Bluetooth exercises and the worksheet. Repeated questions are **merged into one entry**, and the entry lists every paper that asked it. Each topic ends with the most likely questions that come from your notes but have not been asked yet.

**Key**

- ★★★ asked 3–4 times · ★★ asked twice · ★ asked once · ◇ not yet asked, but likely (from the notes)
- **Asked in** shows the paper and marks, e.g. *Exam 1(a) [6]*.
- **SUPPLEMENTARY** means the answer is not in your lecture notes (all of the routing material).
- Answers are written as **exam answers**. Expand each bullet into a full sentence in the exam, at roughly one point per 1–2 marks.

**Topics in order of exam weight** (the exam paper): Routing 30 · Bluetooth 20 · WLAN 18 · Error coding 10 · Wireless channel, spread spectrum, OFDMA, CDMA 20 · Capacity 2.

# 1. Wireless Channel and Propagation

## ★★★ What are the differences between path loss, multipath propagation and interference from other sources?

*Asked in: Exam 1(a) [6] · T1A Q3 [9] · T1B Q4 [12]*

| | Path loss | Multipath propagation | Interference |
|---|---|---|---|
| What | Drop in signal power from Tx to Rx | The **same** signal arrives by several paths with different delays and phases | Unwanted signals from **other** transmitters |
| Cause | Energy spreads over a growing sphere: **$Pt/Pr = (4\pi d/\lambda)^2$**; absorption, obstruction | **Reflection, diffraction, scattering** off buildings, walls, ground, small objects | **Co-channel** (same channel: other WLANs, microwave ovens at 2.4 GHz), **adjacent-channel**, harmonics |
| Depends on | Distance and frequency (path-loss exponent n: 2 free space, 2.7–3.5 urban, 4–6 indoors) | Environment and **movement** | Other users' activity |
| Behaviour | **Predictable**, smooth | **Random, time-varying**: fading (20–30 dB dips), **ISI**, Doppler; strong but poor-quality signal | Random, bursty; lowers SINR |
| Effect | Weak signal, low SNR, limited range | Fading, ISI, distortion, higher BER | Bit errors, blocked transmissions |
| Fix | More power, antenna gain, **more APs** | **Equalization, diversity/MIMO, OFDM, spread spectrum**, lower data rate | **Channel planning**, spread spectrum, directional antennas, remove the source |

**For extra marks**: indoors, path loss grows about **20 dB per 100 ft**. Multipath is worst **indoors**. Interference acts **inward** (others affect us: co-channel) or **outward** (we affect others: adjacent channel).

## ★★ Explain the three basic propagation mechanisms in a mobile communication system.

*Asked in: T1A Q1 [12] · T1B Q3 [12]*

1. **Reflection**: occurs when the wave hits a surface **large compared with the wavelength** (ground, walls, buildings). The **ground-reflected wave can cancel the line-of-sight wave**, and many reflected copies add **constructively or destructively**.
2. **Diffraction**: occurs at the **sharp edge** of an impenetrable body that is large compared with λ. The edge acts as a **new source**, and **secondary waves** reach the **shadow** area, so a signal is received **with no LOS**.
3. **Scattering**: occurs when obstacles are **on the order of λ or smaller** and there are many of them (lamp posts, signs, foliage). The signal splits into **many weaker signals** and is **hard to predict**.

- **With LOS**, reflection dominates. **Without LOS** (urban street level), **diffraction and scattering** are the main means of reception. Together the three cause **multipath**.

## ★ What are the main problems of signal propagation? Why do radio waves not always follow a straight line? Why is reflection both useful and harmful?

*Asked in: T1A Q2 [12]*

- **Problems**: attenuation/free-space loss, noise (thermal, intermodulation, crosstalk, impulse), atmospheric absorption, multipath, fading, ISI, refraction, interference, shared medium and high BER.
- **Not straight**: **refraction** (speed changes with the density of the medium, so the wave bends; the ionosphere returns sky waves), **ground waves** follow the Earth's curvature, and **diffraction/scattering/reflection** redirect the signal.
- **Reflection is useful**: it reaches receivers **without line of sight**, and the extra copies can be **combined by diversity/MIMO**.
- **Reflection is harmful**: delayed, phase-shifted copies cause **destructive interference (fading)**, **ISI** and distortion.

## ★★ A wireless receiving site has occasional loss of signal due to interference. Discuss the steps you would take to correct it.

*Asked in: T1A Q12 [10] · T2-2025 Q6 [10]*

1. **Characterise** the problem: when, where, which channel. "Occasional" points to an intermittent source.
2. **Survey the spectrum** with a spectrum analyser or site-survey tool.
3. **Locate the source**: microwave ovens (2.4 GHz), cordless phones, Bluetooth, neighbouring APs, faulty equipment.
4. **Remove, relocate or shield** the source, or increase its separation from the receiver.
5. **Change the channel** to a clear, non-overlapping one (1, 6 or 11), or **move to 5 GHz**.
6. **Improve the antenna**: placement and height for LOS, **directional antennas**, **diversity**.
7. **Raise the SINR**: more wanted-signal power or antenna gain (within the legal limit).
8. **Use robust techniques**: **spread spectrum/adaptive hopping, FEC, AMC (lower data rate)**, ARQ.
9. **Add APs** or reduce cell size; coordinate channels with neighbouring networks.
10. **Verify and monitor** after each change.

## ★ Explain why many WiFi networks can have a high error rate caused by co-channel interference.

*Asked in: T1A Q13 [4]*

- The admin chooses each AP's channel, and **neighbouring APs often choose the same one**.
- 2.4 GHz has only **3 non-overlapping channels** (1, 6, 11) and is shared with ovens, Bluetooth and phones.
- ISM users have **no regulatory protection**, and co-channel signals add directly to the wanted signal, lowering the SINR.
- **Hidden stations** in other BSSs on the same channel cause collisions that CSMA/CA can't prevent. Result: lost frames, retransmissions, lower throughput.

## ★ A transmitter operates at 850 MHz with a power of 125 mW; the received power is 1 µW. What is the distance?

*Asked in: T1A Q14 [6] (= Problem 2.7 in your notes)*

1. Lp = Pt/Pr = 125 × $10^{-3}$ / 1 × $10^{-6}$ = **125 × $10^{3}$**
2. λ = c/f = 3 × $10^{8}$ / 850 × $10^{6}$ = **0.3529 m**
3. 125 000 = $(4\pi d/0.3529)^{2}$ → √125 000 = **353.55**
4. d = 353.55 × 0.3529 / 4π = **9.93 m**

*(Your notes print "125 × $10^{-3}$" in step 3. It must be **125 × $10^{3}$**.)*

## ◇ Likely questions from the notes

- **Three propagation modes?** Ground wave (< 3 MHz, follows the Earth, AM radio); sky wave (3–30 MHz, ionosphere, thousands of km); line of sight (> 30 MHz, satellite, most wireless).
- **Four types of noise? Thermal** (electron agitation, white, cannot be eliminated); **intermodulation** (sum and difference frequencies); **crosstalk** (unwanted coupling); **impulse** (spikes from lightning or faults; the main source of digital errors).
- **Types of fading? Large-scale** (shadowing, distance) vs **small-scale** (half-wavelength). **Fast** (coherence time Tc not ≫ bit time) vs **slow** (Tc ≫ Tb). **Flat** (coherence bandwidth ≫ signal bandwidth; all frequencies fade together) vs **frequency-selective** (only part of the band fades; needs an equalizer). **Rayleigh** (no LOS) vs **Rician** (LOS + scatter).
- **What are ISM bands?** Licence-free bands (915 MHz, 2.45 GHz, 5.8 GHz, 57–64 GHz), with power limits and spread spectrum required. **No protection** from interference.
- **Why is spectrum underused, and the fix?** Licensing. Dynamic spectrum access + cognitive radio (secondary users use idle primary spectrum). South Africa's regulator is **ICASA**.

# 2. Channel Correction, Capacity and MIMO

## ★★ As a mobile node moves farther from a base station, what two actions could the BS take so that frame loss does not increase?

*Asked in: Exam 5(a) [2] · T1A Q7 [4]*

1. **Increase transmit power**, to offset path loss ($\propto d^2$) and keep the SNR up.
2. **Reduce the data rate**: use more robust modulation and stronger coding (**AMC / rate adaptation**), which needs a lower SNR. *(Also valid: beamform toward the node.)*

## ★ Discuss three methods to increase the capacity of a wireless communication system.

*Asked in: T1B Q5 [12]*

1. **Frequency reuse / small cells (network densification, SDMA)**: reuse frequencies in cells far enough apart; femto- and picocells. Sectorization and beamforming allow tighter reuse (**7-cell → 4-cell pattern nearly doubles capacity**) and intracell reuse.
2. **MIMO**: **SU-MIMO** sends parallel streams (higher rate in the same bandwidth); **MU-MIMO** beams to several users at once (more users per cell); diversity and beamforming add quality and coverage.
3. **Bandwidth expansion**: **carrier aggregation** (20 → 40/80/160 MHz; LTE combines carriers) and **mmWave (30–300 GHz)**, which offers wide channels but suffers from obstructions. *(Other valid choices: CDMA (4–5× the capacity of other digital systems), OFDMA, AMC.)*

## ★ What distinguishes single-user MIMO from multiple-user MIMO?

*Asked in: T1B Q9 [8]*

| | SU-MIMO | MU-MIMO |
|---|---|---|
| Who | All streams to **one user** | Beams to **several users simultaneously** |
| How | **Multilayer**: parallel streams on the same frequency and time | **Beamforming / SDMA** |
| Goal | **Higher data rate** for that user | **Higher capacity** (more users per cell) |
| Needs | Multiple antennas on the user device | Many antennas at the BS + knowledge of each user's channel |

## ◇ Likely questions from the notes

- **Why is backward error correction (ARQ) impractical in wireless?** In satellite links the delay is too long. In mobile links the error rates are so high that the retransmission also fails. So **FEC** is used (total bits/data bits = 2–3).
- **What is adaptive equalization?** It combats **ISI** by gathering dispersed symbol energy back into its time interval, using analog circuits or DSP.
- **Types of diversity? Space** (spaced antennas), **frequency** (OFDM, spread spectrum), **time** (interleaving, Rake receiver). Signals are used by **selection** or **combining**.

- **Four uses of MIMO?** Diversity (performance), beamforming (coverage), MU-MIMO (capacity), multilayer/SU-MIMO (data rate).
- **AMC?** Adapts modulation and coding to the channel hundreds of times per second. Needs **channel measurement** (pilots, loss rates) and **signalling** between Tx and Rx.
- **Three bandwidth expansion approaches?** Carrier aggregation, frequency reuse, mmWave.

# 3. Multiple Access and CDMA

## ★★★ Describe CDMA and explain why it is difficult to jam.

*Asked in: Exam 1(d) [4] · T1A Q6 [4] · T1B Q7 [10]*

**Description**

- **All users share the entire bandwidth all the time**, each separated by a **unique pseudorandom (orthogonal) code**.
- Data are **spread** by the code at a much higher **chip rate**. E.g. IS-95: 1.2288 Mchip/s, 1.25 MHz channel, up to 64 users.
- The signal sits **below the noise level**. The receiver's **correlator despreads** the wanted signal, which passes a narrowband filter; other users remain noise.
- **Power control** keeps everyone at minimum power. **Advantages**: 4–5× the capacity of other digital systems, **soft handoff**, simple frequency planning, privacy.

**Why it is hard to jam**

1. Energy is **spread over a wide band at low power density**, so a jammer must cover the whole band.
2. **Despreading spreads the jammer** (it lacks the code), and the narrowband filter removes most of it. This is the **processing gain** (≈ chip rate / bit rate).
3. The **code is unknown** to the jammer, so it cannot target the signal.
4. Spread spectrum was **designed** to suppress interference and resist interception; it was used in **WWII to foil enemy jamming**.

## ◇ Likely questions from the notes

- **FDMA**: frequency bands with **guard bands**; one user per channel 100% of the time; simple filters, no equalizer; wastes idle channels and guard bands. **N = (Bt − 2Bg)/Bc** (e.g. (25 000 − 200)/200 = 124 channels).
- **TDMA**: one frequency, time slots, **bursts**; needs **synchronisation and equalization**; better spectrum use, longer battery life, simple handoff, no duplexer; high peak power. **Users = channels × slots**.
- **SDMA**: separate users by **space** (beamforming, sectorization, frequency reuse). Benefits: range extension, interference suppression, multipath mitigation, capacity, compatibility.
- **Comparison of SDMA, TDMA, FDMA, CDMA**: separation by directed antennas / time sync / frequency filtering / codes + special receivers.
- **Generations**: 1G analog FDMA (2 kbps) → 2G digital GSM TDMA (64 kbps, SMS) → 3G CDMA/UMTS (2 Mbps, video calls) → 4G all-IP LTE (**OFDMA**, 100 Mbps–1 Gbps, MAGIC) → 5G (> 1 Gbps). *(The notes' table wrongly lists CDMA for 4G and 5G.)*

# 4. Spread Spectrum

## ★★★ Explain the spread spectrum principles of (i) FHSS and (ii) DSSS.

*Asked in: Exam 1(b) [4] · T1A Q4 [10] · T1B Q6 [10]*

**Principle**: expand bandwidth B to **Bss ≫ B** (redundancy), using a **pseudorandom spreading code independent of the signal**. The goals are **privacy and anti-jamming**, not bandwidth efficiency.

**(i) FHSS**

- A **PN code generator** produces a k-bit pattern each **hopping period**; a **frequency table** maps it to one of **$M = 2^k$** carriers; a **frequency synthesizer** generates the carrier, which the data modulate.
- *Example*: M = 8, k = 3; the pattern 101, 111, 001, … selects 700, 900, 300 kHz, …; the sequence repeats after 8 hops.
- **Privacy**: an intruder catches only small pieces. **Anti-jam**: a jammer hits **one hop**, not the whole transmission. **M users** can share Bss (like FDM that changes every hop).

**(ii) DSSS**

- **Each data bit is replaced by n chips**; chip rate = **n × bit rate**, so the bandwidth grows n times.
- *Example*: the **Barker code, n = 11: 10110111000** (wireless LANs). A data 1 sends the code; a data 0 sends the inverted code.
- **Privacy** without the code; **immunity to interference**. Sharing the band requires **orthogonal codes (CDMA)**.

## ★ (Bluetooth exercise 15) How does FH-CDMA differ from DS-CDMA?

- **FH-CDMA**: users are separated by **different hop sequences**; occasional collisions on the same frequency (Bluetooth scatternets).
- **DS-CDMA**: users transmit **simultaneously over the whole band** with **different chip codes** and are separated by correlation (IS-95).

# 5. OFDM and OFDMA

## ★★ What are the benefits of each OFDMA approach: (i) adjacent, (ii) regularly spaced, (iii) randomly spaced subcarriers?

*Asked in: Exam 1(c) [6] · T1A Q5 [9]*

1. **Adjacent** (contiguous block): **all SINRs are about equal**, so the scheduler can **pick the best block for each user**, optimising efficiency, requirements, priority and fairness. It needs full-band channel knowledge (pilots, prediction). Used in LTE and WiMAX.
2. **Regularly spaced** (distributed/comb): **frequency diversity**, so enough good subcarriers whichever are chosen; **low channel-estimation burden**. Used in LTE.
3. **Randomly spaced**: the same **diversity**, **plus reduced adjacent-cell interference**. Used in WiMAX.

## ★ Describe the difference between OFDM and OFDMA.

*Asked in: T1B Q8 [10]*

- **OFDM**: a **modulation technique**. A stream of R bps is split into **N streams of R/N** on **orthogonal subcarriers** (spacing a multiple of 1/T; LTE uses 15 kHz). The spectra **overlap without interference** (≈ 6× FDM in the notes' example). Long symbols (N·Ts) **defeat ISI**, and frequency-selective fading hits only a few subcarriers. **One user uses all subcarriers.**
- **OFDMA**: a **multiple access scheme** = **FDMA + TDMA** on OFDM. **Users get subsets of subcarriers (subchannels / resource blocks of 12) at different times**, with opportunistic scheduling. In the uplink a user concentrates power on its own subchannels. Used in the **LTE downlink** and WiMAX.
- **Not all OFDM uses OFDMA**: 802.11a/g use OFDM with **CSMA** for access.

## ◇ Likely questions from the notes

- **Fundamental requirement of OFDM? Orthogonality**: the average of $s_1 \cdot s_2$ over a bit time = 0; subcarriers are multiples of fb = 1/T.
- **Main strengths?** Robust to frequency-selective fading, no ISI, often no equalizer, spectral efficiency.
- **Main problems? PAPR** (high peaks need expensive linear amplifiers) and **inter-carrier interference** (frequency offset, Doppler).
- **Purpose of SC-FDMA?** An extra **DFT** gives a single-carrier-like signal with **lower PAPR**: better battery life. Used on the **LTE uplink**; one user at a time.
- **Proportional fairness?** Schedule by $\mathbf{r_i / \bar{r}_i}$ (rate available now ÷ the user's own average), balancing efficiency against fairness.

# 6. Error Detection and Correction

## ★★ Hamming code: how many check bits, and what is the codeword?

*Asked in: Exam 2(a) [4] (data 10101111) · T2-2026 Q2 [8] (data 11001010)*

**Method**

1. **2^r ≥ m + r + 1**. For m = 8: r = 3 gives 8 < 12 ✘; r = 4 gives 16 ≥ 13 ✔ → **4 check bits, 12-bit codeword**.
2. Number positions 1–12 from the left. Check bits go in **1, 2, 4, 8**; data fill 3, 5, 6, 7, 9, 10, 11, 12.
3. **P1** covers 3, 5, 7, 9, 11 · **P2** covers 3, 6, 7, 10, 11 · **P4** covers 5, 6, 7, 12 · **P8** covers 9, 10, 11, 12. Even parity: each check bit makes its group's count of 1s even.

| Data | P1 | P2 | P4 | P8 | Codeword |
|---|---|---|---|---|---|
| 10101111 | 1 | 0 | 0 | 0 | **1010 0100 1111** |
| 11001010 | 0 | 0 | 1 | 0 | **0011 1000 1010** |

- **Correcting an error**: recompute the four checks and read them as **C8C4C2C1**; that number is the position of the bad bit.

## ★★★ CRC: find the transmitted bit string.

*Asked in: Exam 2(b) [4] · T1B Q2 [16] · T2-2026 Q3 [6]*

**Method**: generator bits from the polynomial → **append (degree) zeros** → **XOR-divide** → the remainder is the CRC → transmit **data + CRC**. The receiver divides the whole frame: **remainder 0 = OK**, non-zero = error.

| Question | G | Appended | XOR steps | CRC | Transmit |
|---|---|---|---|---|---|
| Exam 2(b): 101110 | 1001 | 101110**000** | 1011→0010, 1010→0011, 1100→0101, 1010→0011 | **011** | **101110011** |
| T2-2026 Q3: 110101 | 1011 ($x^3+x+1$) | 110101**000** | 1101→0110, 1100→0111, 1111→0100, 1000→0011, 1100→0111 | **111** | **110101111** |
| T1B Q2: 10011010 | 1101 ($x^3+x^2+1$) | 10011010**000** | 1001→0100, 1001→0100, 1000→0101, 1011→0110, 1100→0001, 1000→0101 | **101** | **10011010101** |

- **T1B Q2(b)** (leftmost bit flipped): received **00011010101**, remainder **101 ≠ 0**, so an error is detected. A valid frame is always exactly divisible by G.
- **Why modulo-2?** Addition and subtraction become **XOR** with no carries or borrows, giving simple hardware (a shift register + XOR gates) and the mathematical basis of polynomials over GF(2).

## ★★ Internet checksum (4-bit words).

*Asked in: Exam 2(c) [2] · T2-2026 Q4 [4]*

**Method**: add the words; **wrap any carry** around (add it back in); the **checksum = complement** of the sum. The receiver adds everything, including the checksum, and gets **1111** if there is no error.

| Message | Running sums (after wrap) | Sum | **Checksum** |
|---|---|---|---|
| 1001 1100 1010 0011 (Exam) | 0110 → 0001 → 0100 | 0100 | **1011** |
| 1010 1100 0110 1001 (T2-2026) | 0111 → 1101 → 0111 | 0111 | **1000** (receiver: 0111 + 1000 = **1111**) |

## ★★ Single-bit vs burst errors, and how appropriate parity is.

*Asked in: T1B Q1 [10] · T2-2026 Q5 [3] + Q6 [5]*

- **Single-bit error**: only **one bit** changes. **Burst error**: **two or more bits** change. **Burst length** runs from the **first to the last corrupted bit**, inclusive (bits in between may be correct). Bursts dominate in wireless (impulse noise, fading).
- **Parity and single-bit errors**: **always detected**, because one flip makes the count of 1s odd. It is cheap (1 bit), so it suits isolated errors.
- **Parity and burst errors**: **not guaranteed**. Only an **odd** number of flips is caught; any **even** number passes unnoticed (about half of all bursts escape). So parity is **inappropriate for bursts**; use CRC (detects every burst of length ≤ r) or 2-D parity.
- **Can parity correct an error? No.** It cannot tell **which** bit changed.

# 7. Wireless LANs (IEEE 802.11)

## ★★ Use Figure 2 (A, B, C) to explain the hidden node problem. What happens when A and C send simultaneously?

*Asked in: Exam 5(b) [4] · T1A Q8 [4]; also T2-2025 Q4 [10]*

- **B is in range of both A and C**, but **A and C cannot hear each other**. A sending to B is inaudible to C, so C senses the channel as idle.
- **If A and C transmit at the same time**: both frames **collide at B**, which decodes neither. Wireless stations **can't detect the collision**, so both send their whole frames. There are **no ACKs**, both **back off and retransmit**, and **capacity is lost**.
- **Fix**: **RTS/CTS**. B's **CTS** reaches C, which defers.

## ★★ Use Figure 2 to explain how CSMA/CA works in 802.11 (A sends to B).

*Asked in: Exam 5(c) [4] · T1A Q9 [4]*

1. A checks its **NAV** and senses the channel. When idle, it waits **DIFS** and then a **random backoff** (the countdown freezes while the channel is busy).
2. A sends an **RTS** carrying the duration of the whole exchange.
3. After **SIFS**, B sends a **CTS** with the duration. **C hears the CTS and sets its NAV**, so the hidden node stays silent.
4. A sends **DATA** after SIFS; B sends an **ACK** after SIFS.
5. **No CTS or no ACK** means A assumes a collision, backs off and retries.

- **IFS priority**: **SIFS < PIFS < DIFS** (immediate responses < PCF polling < ordinary traffic).

## ★★ Why are acknowledgements used in 802.11 but not in wired Ethernet?

*Asked in: Exam 5(d) [2] · T1A Q10 [4]*

- **Ethernet (CSMA/CD)**: collisions are **detected**, so **no collision means implicit success**, and the cable has a low error rate.
- **802.11**: stations **cannot detect collisions** (they can't send and receive at once, there are hidden stations, and fading) and wireless frames are often lost to **noise, interference and fading**. The sender has **no other way to know the frame arrived**, so it needs an **explicit ACK**, with retransmission when none comes.

## ★★★ Two ISPs in a café, each with its own AP and IP block. (i) Both APs on channel 11? (ii) One on channel 1 and one on channel 11?

*Asked in: Exam 5(e) [4+4] · T1A Q11 [4+4] · T2-2025 Q10 [10+10]*

**(i) Both on channel 11**

- **No breakdown.** The two BSSs share **one collision domain**. CSMA/CA is distributed, so every station senses the channel and NAV and **defers to stations of both ISPs** (taking turns). IP blocks don't matter at the MAC layer.

- **Simultaneous transmissions** (same backoff slot, or hidden from each other) **collide**: no ACK, both **back off** (doubling the window) and **retransmit**.
- Result: more collisions, and **each ISP gets about half the throughput**.

**(ii) Channels 1 and 11**

- These are **non-overlapping channels** (50 MHz apart; each signal is about 22 MHz wide), so the BSSs are **independent**.
- Stations of different ISPs **don't hear or defer to each other**. **Simultaneous transmissions succeed** without collision.
- **Each ISP gets the full channel capacity**: aggregate throughput roughly doubles.

## ★ Compare wired and wireless LANs. What unique concerns must a WLAN designer address?

*Asked in: T2-2025 Q5 [10]*

| | Wired LAN | Wireless LAN |
|---|---|---|
| Medium | Wires | **Air, broadcast, shared** |
| Hosts | Fixed attachment | **Move freely** |
| Isolated LAN | Link-layer switch | **Ad hoc** (no AP) |
| To other networks | Router | **Access point** (joins wireless to wired) |
| Access | CSMA/CD | **CSMA/CA** |
| Channel | Clean | Attenuation, interference, multipath, **high error rate** |
| Moving between them | Change the NICs and swap the switch for an AP: **MAC addresses change, IP addresses stay the same** | |

**Unique concerns**: coverage and path loss; interference in the crowded ISM band (channel plan); multipath and fading; high error rate (ACKs, fragmentation); **hidden and exposed stations**; mobility and roaming; battery and power saving; **security and privacy**; licence-free power limits; QoS.

## ◇ Likely questions from the notes

- **Why can't WLANs use CSMA/CD?** (1) They can't send and receive at once (costly hardware, battery). (2) **Hidden stations** mean collisions go undetected. (3) **Fading** over distance hides collisions.
- **Exposed station problem?** C hears A sending to B and needlessly defers, although its own transmission to D wouldn't interfere. **RTS/CTS cannot fix it.**
- **NAV?** A timer set from the RTS/CTS duration field: **virtual carrier sense**.
- **PCF?** Optional, infrastructure-only **AP polling**; uses **PIFS** for priority over DCF; for time-sensitive traffic. If the AP fails, the system goes down.
- **BSS / ESS / SSID?** BSS = stations + optional AP (ad hoc if there is no AP). ESS = several BSSs joined by a distribution system. SSID = the network name. **Association**: scan for beacons, then request and response.

- **802.11 physical layers?** FHSS and DSSS (1–2 Mbps); infrared (PPM); **a** (OFDM, 5 GHz, 54 Mbps); **b** (HR-DSSS, 11 Mbps); **g** (OFDM, 2.4 GHz, 54 Mbps); **n** (MIMO, 600 Mbps). 2.4 GHz is more crowded; 5 GHz has shorter range but less interference. **Rate adaptation.**
- **Fragmentation?** On a noisy channel it is cheaper to resend a small frame than a large one.

# 8. Bluetooth and IEEE 802.15

## ★★★ Bluetooth profiles add significant complexity. How can they be an advantage to applications?

*Asked in: Exam 4(c) [4] · T2-2025 Q7 [10] · T2-2026 Q7 [4] · BT exercise 19*

1. **Interoperability**: a profile is a *"standard of interoperability so that products from different vendors... will work together"*. Any compliant headset works with any compliant phone.
2. **A ready-made recipe**: each profile states which protocols and features are **mandatory, optional or not applicable** for its usage model, so developers don't have to reinterpret the 2500-page core specification.
3. **Reuse**: profiles build on each other. E.g. the **File Transfer Profile** uses OBEX over RFCOMM and L2CAP, and depends on the Generic Object Exchange, Serial Port and **Generic Access** profiles.
4. **Only what's needed**: devices implement just their profiles, which makes them **cheaper, smaller and lower-power**.
5. **Consistent and plug-and-play**: all devices support the **Generic Access Profile** (discovery, connection, security), and **SDP** lets them find each other's services.

## ★★★ What are the differences between a Bluetooth master and an 802.11 base station / access point?

*Asked in: Exam 4(a) [4] · T2-2026 Q8 [4] · BT exercise 8*

- **Medium access**: the **master polls**. It sets the hop sequence and timing, and slaves transmit **only when addressed** (FH-TDD-TDMA), so there is **no contention**. In **802.11**, stations **contend with CSMA/CA**, collisions are possible, and polling (PCF) is optional.
- **Connecting to other networks**: the **AP links to other networks** (it bridges the BSS to the wired system or Internet) and relays all frames. The **master only sets up the piconet and relays traffic inside it**; there is no infrastructure gateway.
- **Scale**: the master serves **≤ 7 active slaves** over about 10 m. An AP serves many stations over tens to hundreds of metres.
- **Role**: any Bluetooth device can be master (a dynamic role); an AP is a dedicated device.

## ★★★ A Bluetooth device can be in two piconets. Can one device be the controller (master) of both at the same time?

*Asked in: Exam 4(d) [4] · T2-2025 Q9 [5] · BT exercise 9*

- **No.** A piconet is **defined by its master**: the **hopping sequence and timing** come from the **master's own address and clock**.
- If one device mastered both, they would use the **same hop sequence and timing**, so they would be **one piconet**, not two (and limited to 7 slaves).
- A device **can** be **slave + slave** or **master + slave**, time-sharing between the two piconets. That is how a scatternet works.

## ★★★ Scatternet of two piconets, no CSMA, 80 frequencies, always busy. Probability of collision (a success needs both the poll and the reply)?

*Asked in: Exam 4(e) [4] · T2-2025 Q8 [10] · BT exercise 20*

1. The piconets hop **independently**, so P(same frequency in a slot) = **1/80**.
2. P(a slot is clear) = **79/80**.
3. Poll + reply = **2 independent slots**, so P(success) = $(79/80)^2$ = **0.9752**.
4. **P(collision) = 1 − $(79/80)^2$ ≈ 0.025 = 2.5%** (it is 1/80 = 1.25% per single packet).

## ★★ Does the hidden terminal problem occur in Bluetooth?

*Asked in: Exam 4(b) [4] · T2-2025 Q4 [10]*

- **No, not within a piconet.** The hidden terminal problem needs **contention**: stations transmitting on their own after sensing an idle channel.
- In Bluetooth the **master polls**: a slave sends **only in the slot right after it is addressed**, and voice (SCO) slots are reserved. **Two slaves never start transmitting together**, so there is nothing to collide at the master.
- Collisions **between piconets** (the same hop in the same slot) are an FH-CDMA effect handled by FEC and ARQ, not a hidden-terminal problem.

## ★ Bluetooth exercises (quick answers)

| # | Question | Answer |
|---|---|---|
| 1 | Why a WPAN, not a WLAN? | ~10 m personal range, low power and cost, cable replacement, ≤ 8 active devices |
| 2 | Other name for Bluetooth Smart; which version? | **Bluetooth Low Energy (BLE)**, **v4.0** |
| 3 | How does 3.0 reach 24 Mbps? | **AMP**: uses an **802.11 (Wi-Fi) link** for data; Bluetooth handles discovery and setup |
| 4 | Piconet vs scatternet? | Piconet = 1 master + ≤ 7 slaves on one hop sequence. Scatternet = overlapping piconets that share area and bandwidth (FH-CDMA) |
| 5 | More than 8 stations? | Not active: the **3-bit AM_ADDR** allows 7 slaves + the master (more can be parked) |
| 6 | Master–slave relationship? | The master sets the hop sequence and timing; slaves talk only to the master, and only when polled |
| 7 | Link types? | **SCO** (reserved, symmetric, no retransmission, voice); **eSCO** (reserved, can be asymmetric, with retransmission); **ACL** (point-to-multipoint, packet-switched, retransmission) |
| 10 | Role of the radio layer? | The air interface: 2.4 GHz, 79 × 1 MHz channels, hopping, modulation, power classes |
| 11 | Actual bandwidth used? | **1 MHz** at a time; hops across ~79 MHz |
| 12 | Spread spectrum technique? | **FHSS, 1600 hops/s** (625 µs slots) |
| 13 | Modulation? | **GFSK** (1 = +deviation, 0 = −deviation, 1 Mbps); EDR uses π/4-DQPSK (2 Mbps) and 8DPSK (3 Mbps) |

| # | Question | Answer |
|---|---|---|
| 14 | How are FH and TDD combined? | Each 625 µs slot is on the next hop frequency, and the direction alternates master→slave then slave→master |
| 16 | Interference from microwave ovens? | **Yes**: ovens use ~2.45 GHz. The effect is limited by hopping, adaptive hopping, and FEC/ARQ |
| 17 | Baseband MAC? | **TDMA with master polling (FH-TDD-TDMA)** |
| 18 | Role of L2CAP? | **Adapts upper-layer protocols to the baseband**; connectionless and connection-oriented services |

## ◇ Likely questions from the notes

- **Core protocols? Radio** (air interface) · **Baseband** (connections, addressing, packet format, timing, power control) · **LMP** (link setup, authentication, encryption) · **L2CAP** (adaptation) · **SDP** (service discovery). Plus **RFCOMM** (virtual serial port, RS-232), **TCS BIN** (call control) and adopted protocols (PPP, TCP/UDP/IP, OBEX).
- **Power classes?** Class 1: 100 mW (power control mandatory) · Class 2: 2.5 mW · Class 3: 1 mW.
- **Packet format?** Access code 72 bits · header 54 bits (18 × 3 for FEC: AM_ADDR 3, Type 4, Flow 1, ARQN 1, SEQN 1, HEC 8) · payload 0–2745 bits.
- **Error control? 1/3 FEC** (each bit sent 3 times, majority vote; used on the header), **2/3 FEC** (Hamming (15,10): corrects 1 error, detects 2), **ARQ** (CRC, ACK/NAK, timeout).
- **Purposes of frequency hopping?** (1) Resistance to interference and multipath. (2) Multiple access between piconets. Adaptive hopping uses at least 20 channels.
- **Bluetooth Smart (BLE)?** 40 channels × 2 MHz (3 advertising + 37 data), 10 mW, 150 m, single-mode or dual-mode, aimed at the IoT.
- **802.15.3 PNC vs AP vs master?** The **PNC controls access to time but is not involved in data exchange between devices**; the AP links to other networks; the master relays all piconet traffic.
- **ZigBee device types? Coordinator** (one per network; starts it; holds security keys) · **Router** (relays) · **End device** (can't relay; sleeps; cheapest). Built on 802.15.4, 20–250 kbps, mesh, uses **AODV**.

# 9. Routing and Ad Hoc Networks (SUPPLEMENTARY: no notes provided)

## ★ Distinguish between routing, packet forwarding and switching: purpose and the information used.

*Asked in: T2-2026 Q1 [6]*

| | Purpose | Information used |
|---|---|---|
| **Routing** | Decide the **end-to-end path** across the network (control plane) | **Topology and link costs** from routing protocols, which build the forwarding table |
| **Forwarding** | Move a packet from **input to output link inside a router** | The packet's **destination IP address** + the **forwarding table** (longest-prefix match) |
| **Switching** | Move **frames** to the right port within a LAN (layer 2) | The **destination MAC address** + the **switch table** (self-learned) |

## ★★ Why is routing in ad hoc networks complicated? State and explain the three categories of MANET routing protocols, with examples.

*Asked in: Exam 2(d) [10] · T2-2025 Q1 [12]*

**Why it is complicated**: **mobility** (links keep breaking); **no infrastructure** (every node must route); **limited bandwidth**; **limited battery and CPU**; **unreliable links** (fading, hidden terminals, asymmetric links); multi-hop delay; security.

1. **Proactive (table-driven)**: keep routes to **all** destinations at all times through periodic updates. Routes are available **immediately**, but control overhead is **high and constant**. Examples: **DSDV**, OLSR.
2. **Reactive (on-demand)**: find a route **only when needed**. **Route discovery** floods a request and the destination replies; **route maintenance** repairs breaks. **Low overhead**, but a delay before the first packet. Examples: **AODV** (used by ZigBee), DSR.
3. **Hybrid**: **proactive within a nearby zone, reactive between zones**. Example: **ZRP**.

## ★ Compare and contrast link-state and distance-vector routing.

*Asked in: Exam 3(a) [4]*

- **Same**: both find least-cost paths, build forwarding tables, and adapt to changes.
- **Different**: link-state has **global** knowledge (flood link costs to **all**, run **Dijkstra**, **fast**, no loops, robust; e.g. OSPF). Distance-vector has **local** knowledge (send the vector to **neighbours** only, run **Bellman-Ford**, **slow**, loops, **count-to-infinity**, errors spread; e.g. RIP).

## ★ What is the essential difference between Dijkstra and Bellman-Ford?

*Asked in: Exam 3(b) [4]*

- **Dijkstra**: **global, greedy**. It needs the whole graph, **settles the nearest node permanently each step** (n steps), and requires non-negative costs.

- **Bellman-Ford**: **distributed, iterative**. It uses only **neighbours' estimates**, $d_x(y) = \min_v\{c(x,v) + d_v(y)\}$, and **relaxes repeatedly until convergence**. It handles negative costs, but converges slowly.

## ★★ Apply Dijkstra to Figure 1 (from x, t and w).

*Asked in: T2-2025 Q2 [8] (from x) · Exam 3(c) [4+4] (from t and w)*

**Figure 1 links**: x–z 8, x–y 6, x–v 3, x–w 6, y–z 12, y–v 8, y–t 7, v–t 4, v–u 3, v–w 4, t–u 2, w–u 3. **Method**: draw a table of Step | N' | D(n),p(n). Each step, add the node with the smallest D to N' and update its neighbours with D = min(old, D(new) + cost).

**From x**

| Step | N' | t | u | v | w | y | z |
|---|---|---|---|---|---|---|---|
| 0 | x | ∞ | ∞ | **3,x** | 6,x | 6,x | 8,x |
| 1 | xv | 7,v | **6,v** | | 6,x | 6,x | 8,x |
| 2 | xvu | 7,v | | | **6,x** | 6,x | 8,x |
| 3 | xvuw | 7,v | | | | **6,x** | 8,x |
| 4 | xvuwy | **7,v** | | | | | 8,x |
| 5 | xvuwyt | | | | | | **8,x** |

**Result**: v 3 (x-v) · u 6 (x-v-u) · w 6 (x-w) · y 6 (x-y) · t 7 (x-v-t) · z 8 (x-z)

**From t**

| Step | N' | u | v | w | x | y | z |
|---|---|---|---|---|---|---|---|
| 0 | t | **2,t** | 4,t | ∞ | ∞ | 7,t | ∞ |
| 1 | tu | | **4,t** | 5,u | ∞ | 7,t | ∞ |
| 2 | tuv | | | **5,u** | 7,v | 7,t | ∞ |
| 3 | tuvw | | | | **7,v** | 7,t | ∞ |
| 4 | tuvwx | | | | | **7,t** | 15,x |
| 5 | tuvwxy | | | | | | **15,x** |

**Result**: u 2 (t-u) · v 4 (t-v) · w 5 (t-u-w) · x 7 (t-v-x) · y 7 (t-y) · z 15 (t-v-x-z)

**From w**

| Step | N' | t | u | v | x | y | z |
|---|---|---|---|---|---|---|---|
| 0 | w | ∞ | **3,w** | 4,w | 6,w | ∞ | ∞ |
| 1 | wu | 5,u | | **4,w** | 6,w | ∞ | ∞ |
| 2 | wuv | **5,u** | | | 6,w | 12,v | ∞ |
| 3 | wuvt | | | | **6,w** | 12,v | ∞ |
| 4 | wuvtx | | | | | **12,v** | 14,x |
| 5 | wuvtxy | | | | | | **14,x** |

**Result**: u 3 (w-u) · v 4 (w-v) · t 5 (w-u-t) · x 6 (w-x) · y 12 (w-v-y; **tie** with w-x-y and w-u-t-y) · z 14 (w-x-z)

## ★ Apply Dijkstra from A (T2-2026 figure); give the final costs and routes.

*Asked in: T2-2026 Q9 [6+4]*

**Links**: A–B 2, A–G 6, B–C 7, B–E 2, E–G 1, E–F 2, F–C 3, F–H 2, G–H 4, C–D 3, H–D 2.

| Step | N' | B | C | D | E | F | G | H |
|---|---|---|---|---|---|---|---|---|
| 0 | A | **2,A** | ∞ | ∞ | ∞ | ∞ | 6,A | ∞ |
| 1 | AB | | 9,B | ∞ | **4,B** | ∞ | 6,A | ∞ |
| 2 | ABE | | 9,B | ∞ | | 6,E | **5,E** | ∞ |
| 3 | ABEG | | 9,B | ∞ | | **6,E** | | 9,G |
| 4 | ABEGF | | 9,B | ∞ | | | | **8,F** |
| 5 | ABEGFH | | **9,B** | 10,H | | | | |
| 6 | ABEGFHC | | | **10,H** | | | | |

**Routes**: B **2** (A-B) · C **9** (A-B-C; tie with A-B-E-F-C) · D **10** (A-B-E-F-H-D) · E **4** (A-B-E) · F **6** (A-B-E-F) · (G 5 via E; H 8 via F)

## ★★ What is the count-to-infinity problem? Does it occur if a link cost decreases, or if two unlinked nodes are connected?

*Asked in: Exam 3(d) [4] · T2-2025 Q3 [5]*

- **What it is**: in distance-vector routing, **bad news travels slowly**. After a link cost rises or a link fails, a node adopts a neighbour's **stale** route that actually passes back through itself. This creates a **routing loop**: the estimates creep up step by step towards the true cost, or **towards infinity** if the destination is unreachable.
- *Example*: y–x 4, y–z 1, z–x 50, and the y–x cost changes to 60. y switches to the route via z (cost 6), z then goes via y (7), and so on: **44 iterations**, with packets looping meanwhile. Fixes: **poisoned reverse**, split horizon, a maximum hop count.
- **Link cost decreases**: **No**. Good news travels fast: lower costs propagate quickly and no loop forms.
- **Connecting two unlinked nodes**: **No**. This is the same as a cost falling from ∞ to a finite value, which is good news.

# 10. One-Page Formula and Method Sheet

| Formula / method | Use |
|---|---|
| $T = 1/f \cdot \lambda = c/f$ ($c = 3 \times 10^8$ m/s) | Basics |
| **$C = 2B \log_2 M$** | Nyquist (noiseless channel) |
| **$C = B \log_2(1 + SNR)$**, with SNR = 10^(dB/10) | Shannon (noisy channel); **convert dB first** |
| **$Pt/Pr = (4\pi d/\lambda)^2$** · $Pr = Pt/(4\pi d/\lambda)^2 \cdot d = (\lambda/4\pi)\sqrt{(Pt/Pr)}$ | Free-space power and distance |
| L_dB = 20 log f + 20 log d − 147.56 | Free-space loss in dB |
| SNR = Pr/Pn · Lp = Pt/Pr | Notes' equations 2.2 and 2.3 |
| N = (Bt − 2Bg)/Bc · users = channels × slots | FDMA and TDMA |
| S = N/r, $r = \log_2 L$ | Baud rate vs bit rate |
| **2^r ≥ m + r + 1** | Number of Hamming check bits |
| CRC: append (degree) zeros, XOR-divide, send data + remainder | CRC |
| Checksum: add with wrap-around, then complement; receiver gets 1111 | Internet checksum |
| **$1 - (79/80)^2$** | Scatternet collision probability |
| **$d_x(y) = \min_v\{c(x,v) + d_v(y)\}$** | Bellman-Ford |
| $r_i / \bar{r}_i$ | Proportional fairness |

## Top 10 mistakes

1. Putting a dB value straight into Shannon's formula.
2. Writing $125 \times 10^{-3}$ instead of the path-loss ratio $125 \times 10^3$.
3. Appending the wrong number of CRC zeros (it must equal the generator's degree).
4. Using ordinary subtraction instead of XOR in CRC division.
5. Forgetting the wrap-around carry or the final complement in the checksum.
6. Using 3 check bits instead of 4 for an 8-bit Hamming code.
7. Saying parity corrects errors, or that it catches every burst.
8. Skipping Dijkstra rows, or forgetting to update a node when a cheaper path appears.
9. Saying 802.11 breaks down when both APs use channel 11.
10. Saying one device can be master of two piconets, or that a piconet has 8 slaves (it has 7 active slaves + 1 master).

## Exam technique

**Yes/No questions**: give the answer in the first word, then justify it. **Comparisons**: use a table. **Scenarios**: name the nodes (A, B, C). **Calculations**: write the formula, then each step, then the boxed answer, then a check. **Budget**: one point per 1–2 marks, about one minute per mark.